AdaptHAD: Adaptive One-Step Hybrid Network for Hyperspectral Anomaly Detection

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***Abstract*—Deep learning for anomaly detection is one of the current hot topics in remote sensing. However, most deep learning-based methods require retraining or parameter fine-tuning when detecting different hyperspectral images (HSIs), which increases the complexity of model usage and wastes computing resources. Although a few one-step detection methods exist, they are generally constrained by fixed spatial and spectral dimensions, limiting their flexibility in handling different HSIs. More importantly, this constraint often disrupts the subtle and critical relationships between spectral bands, affecting the comprehensiveness and accuracy of image analysis. To overcome these issues, we propose an adaptive one-step hybrid network for hyperspectral anomaly detection (denoted as AdaptHAD). First, an adaptive sparsity module is constructed to assist in dividing samples in the training phase by utilizing the sparsity property of anomalies in HSIs. Second, a data augmentation method (DAM) that combines spectral information erasure with random noise is proposed to address the small sample problem. Third, a convolutional attention hybrid network (CAN) with a transformer and a convolutional neural network (CNN) dual branches is designed to extract global and local features simultaneously to strengthen the model to identify anomalies and backgrounds. Extensive experiments on real-world HSIs demonstrate the effectiveness of the proposed method. The code is available at https://github.com/ismeXQ/AdaptHAD**

***Index Terms*—Adaptive sparsity, hybrid network, hyperspectral anomaly detection (HAD), one-step detection, unsupervised learning.**

I. Introduction

THE high-spectral resolution of hyperspectral remote sensing allows the characteristic curves to be obtained continuously and finely, which brings favorable conditions for ground material feature analysis. It has been widely used in hyperspectral image (HSI) processing, such as image fusion [1], classification [2], [3], unmixing [4], [5], and anomaly detection [6], [7]. Hyperspectral anomaly detection (HAD) refers to identifying the pixels whose spectral characteristics in the HSI are significantly different from adjacent or global background pixels, without the aid of prior information [8]. It has an important application value in military defense, early forest fire, and offshore oil spill surveillance [9], [10], [11], [12].

HAD methods have been developed over several decades, evolving from statistical modeling to representation learning, and more recently, to deep learning. Statistics-based approaches focused on exploiting the statistical properties of pixels for background modeling and anomaly identification. However, when confronted with complex, high-dimensional backgrounds and nonlinear features, these methods lacked generalization and robustness. Subsequently, representation learning techniques employed nonlinear methods such as low-rank (LR) sparse modeling, dictionary learning, and graph learning to better represent complex backgrounds and identify sparse anomalies. Despite these advances, these methods were sensitive to parameter selection and exhibited limited scalability. In recent years, deep learning technology has become a research hotspot in this field due to its end-to-end feature learning capabilities and superior nonlinear modeling performance, and has further promoted the development of HAD. However, most deep learning-based models require retraining the trained network model or fine-tuning the parameters when detecting different HSIs to achieve the best detection results. Few deep models can be used directly for HAD of any spatial size or spectral dimension. Although the deeper the number of layers of the network model, the higher the accuracy of the model, the greater the demand for computing resources, and the number of training samples. Therefore, it is crucial to design a one-step detection framework and a lightweight detection model.

Fig. 1 illustrates the difference between two-step and one-step detection methods. Specifically, two-step detection requires retraining or parameter fine-tuning to obtain the best detection results before detecting different input HSIs. In contrast, one-step detection refers to directly testing different input HSIs to get the best results without retraining or fine-tuning the parameters.

To overcome the above problems, an adaptive one-step hybrid network for HAD (AdaptHAD) is proposed in this work. This method is not limited to the spatial and spectral dimensions of the input HSI and can achieve one-step detection on different datasets without retraining. First, to

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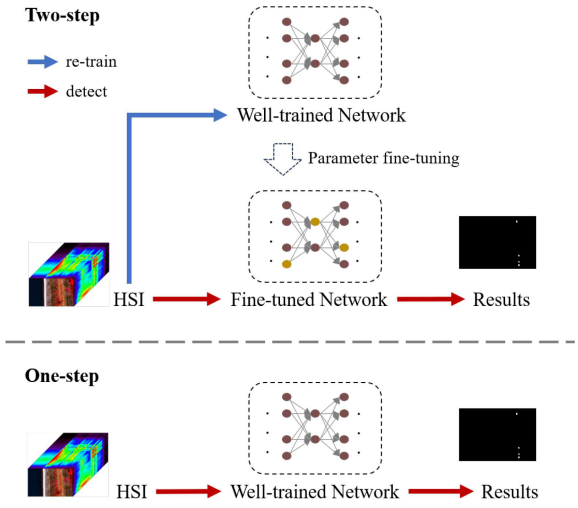


Fig. 1. *Two-step detection:* retraining is required before detection of the HSI. *One-step detection:* direct inference is used to obtain detection results.

ensure the transferability of the AdaptHAD method, we propose an adaptive sparse module (ASM) by utilizing a property common to HAD datasets, i.e., the sparse property of anomalies. The module acts as a sampler in the training phase and divides the input HSI into potential anomaly samples and background samples. Second, the number of anomaly samples in HAD is extremely limited, while deep learning models require a sufficient amount of training samples to capture data features and improve generalization capabilities fully. To address this problem, we propose a data augmentation method (DAM) that erases the spectral information in different degrees and combines it with random noise. The augmented samples obtained through this method preserve the original spectral information, ensuring that the semantics and distribution of the data remain consistent before and after augmentation. Third, we propose a convolutional attention hybrid network (CAN), which consists of two branches, a transformer and a convolutional neural network (CNN). It can not only extract information within the pixel spectrum bands but also capture subtle differences between anomalies and the background.

The main contributions of this article are summarized as follows.

1) We propose an adaptive sparsity module that capitalizes on the anomaly sparsity property common to HAD images. The module acts as a sampler in training to initially classify the input data into potential anomalies and backgrounds, which opens new ways for unsupervised HAD methods in sample selection.
2) To solve the small sample problem, a DAM is proposed. This approach involves erasing spectral information and combining it with random noise. It ensures consistent semantics and distribution before and after data augmentation, while also increasing the diversity among samples. This provides a new method for data augmentation of high-dimensional data.
3) To simultaneously extract inter-band information, as well as the difference information between anomalies and backgrounds, a CAN is proposed. This network model is not limited by the spatial size and spectral dimensions of HSIs.

The remainder of this article is as follows. Section II reviews and analyzes existing research on HAD. The details of the proposed method are described in Section III. Section IV presents a comparative analysis of classical and state-of-the-art methods to verify the effectiveness of the proposed method. Section V is a summary and outlook of this work.

II. Related Work

In the past few decades, research on HAD has advanced substantially and is commonly categorized into three main approaches: statistics-based, representation learning-based, and deep learning-based.

Early statistics-based approaches are mainly represented by the Reed-Xiaoli (RX) anomaly detector and its variants. The RX [13] method is based on the Mahalanobis distance and background statistical characterization via the mean vector and covariance matrix. To better handle complex backgrounds, several variants have been developed, including robust covariance estimation [14], [15], [16], subspace projection methods [17], [18], [19], and kernel-based methods [20], [21], [22], which aim to enhance background modeling and improve the estimation of hyperspectral covariance matrices. In addition, entropy-based anomaly detection methods [23], [24] characterize the difference between anomalies and background through information theory metrics.

Representation learning-based approaches aim to enhance adaptability to complex backgrounds and better capture nonlinear data characteristics in some cases, such as collaborative representation (CR), sparse representation (SR), and LR decomposition [25], [26], [27]. Early studies used matrix decomposition [28], [29] to separate LR background and sparse anomalies, but flattening hyperspectral data into a 2-D matrix destroys its inherent 3-D spatial-spectral structure. To address this problem, tensor-based models [30], [31], [32] were introduced into the LR and sparse framework, which more effectively preserves spatial–spectral information. Subsequent studies further optimized LR sparse methods by integrating spatial-spectral joint features, incorporating prior information, suppressing noise, and simplifying parameter adjustment [33], [34], [35]. Compared with SR-based methods, CR-based algorithms emphasize the overall collaboration between samples. For example, the CR-based anomaly detector (CRD) [25] distinguishes anomaly from background by representing background pixels as a linear combination of neighboring pixels and minimizing the l_2-norm of the representation weights to enhance the collaboration. The subsequent improvement methods also promoted its further development [36], [37]. However, such methods still face challenges such as sensitive parameter settings and the difficulty of separating anomalies from the background in complex scenes.

With the development of artificial intelligence, deep learning has become a research hotspot for HAD because of its ability of nonlinear modeling and automatic feature extraction, breaking through the limitations of traditional methods such as strong reliance on priors and poor adaptability. At present, mainstream methods include those based on CNN [38], [39],

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[40], [41], [42], autoencoder (AE) [43], [44], [45], graph neural network (GNN) [46], [47], [48], generative adversarial network (GAN) [49], [50], Transformer [51], [52], [53], [54], and the latest Mamba architecture [55], [56]. These methods have different focuses in terms of feature representation ability, generalization performance, and computational efficiency. For example, graph AE and Transformer can effectively model global dependencies, but the computational cost is high; CNN and AE are good at reconstructing local features, but have limited adaptability to complex backgrounds; models such as GAN and Mamba use dynamic learning mechanisms to improve generalization ability, but model stability is still a challenge. In addition, some hybrid paradigms that integrate physical modeling and data-driven methods have also attracted attention [34], [57], [58], [59]. In general, existing deep learning-based methods adopt a two-step detection framework that requires retraining or hyperparameter adjustment for specific datasets, which increases deployment complexity and computational overhead. Therefore, a one-step detection strategy has become a more effective solution for HAD.

Some researchers have focused on improving the generalizability for models, and a few one-step deep detection models have been developed. For example, Li et al. [60] proposed a once-only trained anomaly detection method (AETNet), which utilized random masks for data augmentation and combined it with Swin UNet for background reconstruction. The Mahalanobis distance of the network reconstruction output determines the final result. Li et al. [61] proposed an unsupervised transferred direct detection (TDD) model to optimize the HAD task through the spectral deviation relationship and directly output the anomaly detection map. However, the number of bands in the input layer of this model is fixed, which breaks the potential relationship between spectral bands when detecting HSIs with different numbers of bands. Huyan et al. [62] combined a dual-window strategy and transformer to extract relation embeddings of pixels for background reconstruction, and the centers with significant differences in relational embedding features will be considered anomalies (AUD-Net). However, this method requires setting the size of the dual windows according to the spatial dimensions of the HSI. Jin et al. [63] introduced a feature enhancement network (FEN) in conjunction with the RX algorithm for anomaly detection. Despite its effectiveness, the method inherently depends on traditional prior assumptions. Furthermore, to accommodate HSIs with varying spectral dimensionality, distinct models need training to ensure optimal performance. In essence, existing one-step deep detection frameworks have limited adaptability to the variations in spatial and spectral properties of HSIs.

III. Proposed Method

Fig. 2 depicts the functionality of the proposed framework. In this section, we provide a comprehensive overview of the proposed ASM and DAM. Furthermore, we discuss the proposed CAN and the loss function used.

A. Notations

In this work, italic denotes scalars, lowercase bold denotes vectors, and uppercase bold denotes matrices. $\mathbb{R}$ denotes the dimension of the space. Let $\mathbf{X} = \{x_{ij}\}_{i=1,j=1}^{N,D} \in \mathbb{R}^{N\times D}$ be the hyperspectral data, where N is the number of pixels and D is the number of bands. $\mathbf{p}_j = \{x_{ij}\}_{i=1}^{N}$ denotes a vector consisting of the spectral values of N pixels in the jth band.

B. Adaptive Sparse Module

Anomalies have sparse properties in HSIs. Based on this property, we facilitate the separation of anomalies from the background. l_1-norm is commonly used to implement feature selection and sparsity in machine learning and statistical modeling. In this work, we explore the relationship between l_1-norm and l_k-norm use it to preliminarily divide the input batch dataset into potential anomalies and backgrounds during the training phase. The formula is expressed as follows:

$$\mathbf{l}_k = \|\mathbf{X}\|_k = \left(|\mathbf{p}_1|^k + |\mathbf{p}_2|^k + \cdots + |\mathbf{p}_D|^k\right)^{(1/k)} \tag{1}$$

$$\mathbf{s} = \exp\left\{[(1-k)/k] \times \log N\right\} \times (\mathbf{l}_1/\mathbf{l}_k), \quad k \in [1, D-1] \tag{2}$$

where k is the order of the norm. $\|\cdot\|_k$ denotes the k norm notation and $|\cdot|^k$ denotes taking the absolute value of the vector first and then raising the result to the kth power. $\mathbf{s}$ represents the proportion of potential anomalies in the current batch dataset. In the hyperspectral dataset, $\mathbf{l}_k$ is calculated along the spectral dimension, with the range of k being $[1, D-1]$. Specifically, the value of k needs to be adjusted according to the convergence speed of the loss during training.

The proportion of anomalies in the current batch dataset is calculated using (1) and (2) to obtain potential anomaly pixels. Because each batch dataset is different, their $\mathbf{l}_1$ and $\mathbf{l}_k$ values also vary. Consequently, when using (2) to calculate the $\mathbf{s}$ value for different data batches, the value changes dynamically, demonstrating the adaptability of the ASM.

The ASM is shown in Fig. 2. First, the first-order difference is calculated for the input batch data to extract its spatial information, such as the boundary. Second, the l_1-norm of the first-order difference is calculated and arranged in descending order. A smaller l_1 value of the first-order difference indicates less variation between a pixel and its surrounding pixels, suggesting that the pixel is more likely part of the background. Therefore, when combined with the proportion of potential anomalies, the first U pixels in the l_1-norm of the first-order difference can be considered potential anomalies. Specifically, we analyze the k values chosen for this work in Section IV-A1 of this article. Third, the first U pixels of the l_1-norm result are selected as potential anomaly pixels (U is equal to the proportion multiplied by the total number of pixels in the current batch dataset). The remaining pixels are regarded as background pixels.

C. Data Augment Method

Considering that the anomaly and background samples derived from the ASM exhibit insufficient accuracy, and the small number of anomalies fails to meet the requirements for effective model training, we propose a DAM to alleviate these issues. Specifically, we enhance the anomalies by

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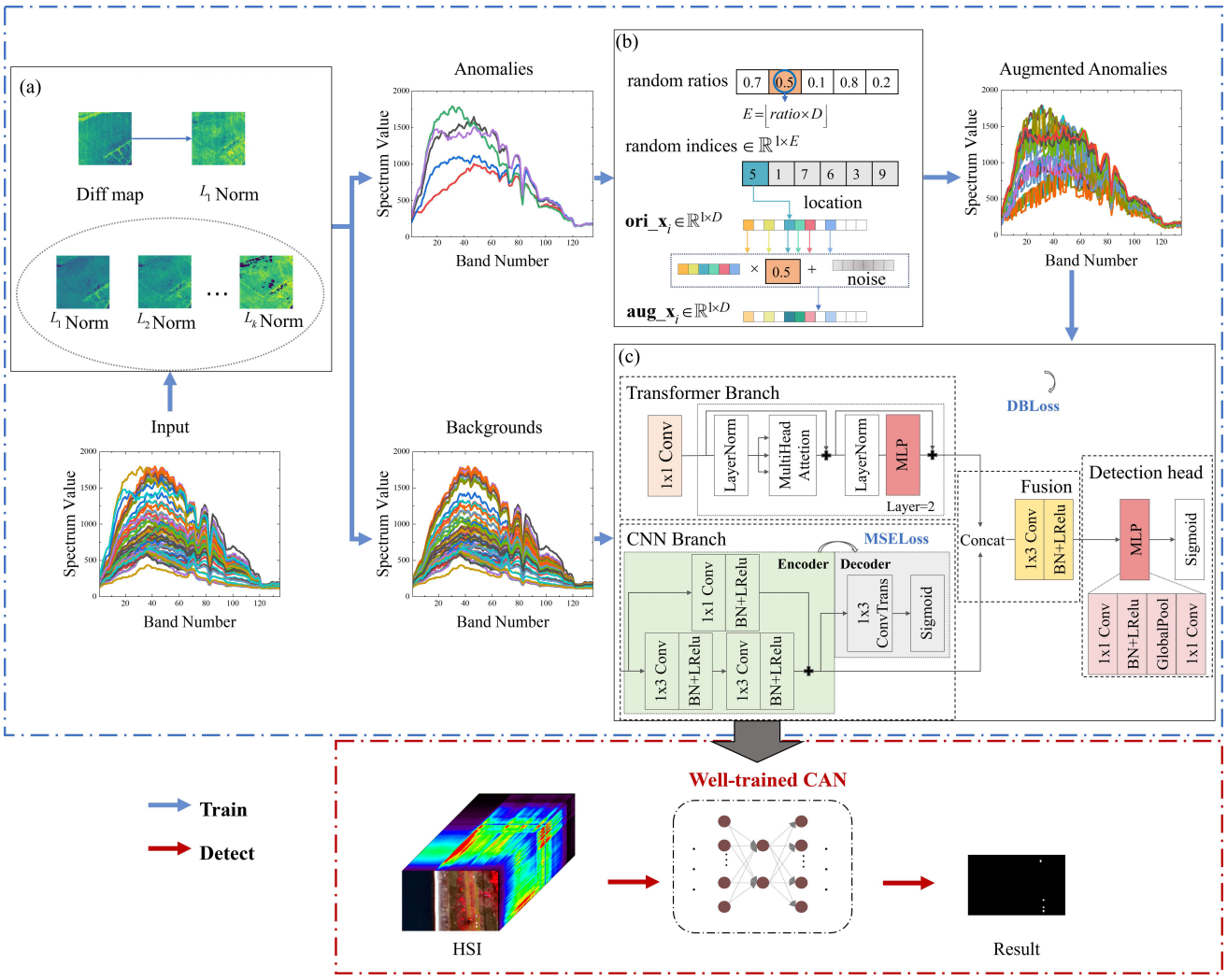


Fig. 2. Illustration of the proposed AdaptHAD. Solid lines with blue arrows represent training, and solid lines with red arrows represent detection. In the training phase, first, the input spectrum is divided into background samples and anomaly samples after (a) ASM. Second, to solve the small sample problem, (b) DAM is applied to the anomaly samples to equalize the number of anomaly samples and background samples. Next, the background samples and the augmented anomaly samples are fed into (c) CAN, which is equipped with the ability to identify anomalies by learning the difference between anomalies and backgrounds. In the detection phase, the HSI is input into the well-trained CAN, and the anomaly detection map is directly outputted.

applying varying degrees of spectral information erasure and introducing random Gaussian noise. This has the following two benefits. First, the background in the anomaly is weakened by adding a noise transformation, thereby reducing the interference of the intersection between the anomaly and the background on the model training. Second, the augmentation preserves the original anomaly information while enhancing the diversity of the anomaly spectrum.

The DAM module in Fig. 2 shows the proposed DAM. The basic idea is to erase the spectral band information to varying degrees while adding random noise to generate new data. It is worth noting that we use Gaussian noise in this work to better model the unknown real noise.

Let $\mathbf{Y} = \{y_{ij}\}_{i=1,j=1}^{N,D} \in \mathbb{R}^{N\times D}$ be the original spectrum and $\mathbf{Z} = \{z_{ij}\}_{i=1,j=1}^{N,D} \in \mathbb{R}^{N\times D}$ be the augmented spectrum, where N is the number of pixels and D is the number of bands. First, a set of random numbers ranging from 0 to 1 (denoted as "spectrum erasure ratio" and "original spectrum retention ratio") is generated through uniform distribution. It can be expressed as $\mathbf{r} = \{r_t\}_{t=1}^{N}$. Second, the number of bands to be erased can be calculated using ratios and the number of bands in the HSI. Taking the tth ratio value r_t as an example, the process can be formulated as

$$E = \lfloor r_t \times D \rfloor \tag{3}$$

where E denotes the number of bands to be erased, and $\lfloor \cdot \rfloor$ represents the floor operation. Third, the positions of the bands to be erased are determined. Specifically, a set of E random integers within the range $(0, D)$ is generated, denoted as $\mathbf{d} = \{d_t\}_{t=1}^{E}$. Finally, we introduce a set of Gaussian random noise $\mathbf{n} = \{n_t\}_{t=1}^{N}$ and generate new spectra through a spectral synthesis strategy. Here, $\mathbf{y}_t = \{y_{\text{it}}\}_{i=1}^{N}$ represents the original spectral values of the N pixels at position d_t, while $\mathbf{z}_t = \{z_{\text{it}}\}_{i=1}^{N}$ denotes the augmented spectrum vector at the same position. The data augment expression is as follows:

$$\mathbf{z}_t = r_t \times \mathbf{y}_t + \mathbf{n}. \tag{4}$$

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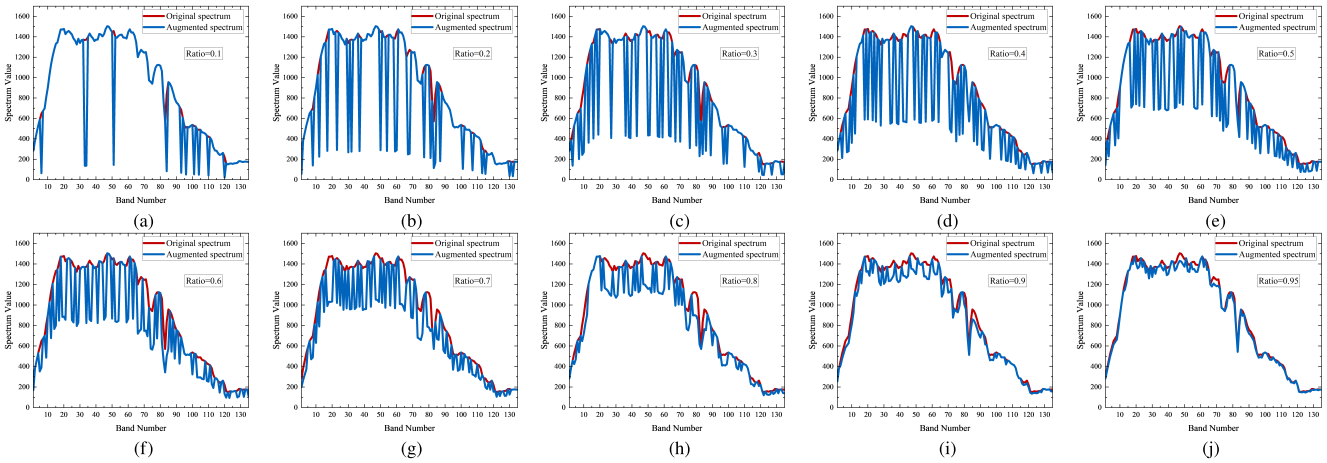


Fig. 3. Visualization of augmented samples under different anomaly ratios. (a) Ratio = 0.1. (b) Ratio = 0.2. (c) Ratio = 0.3. (d) Ratio = 0.4. (e) Ratio = 0.5. (f) Ratio = 0.6. (g) Ratio = 0.7. (h) Ratio = 0.8. (i) Ratio = 0.9. (j) Ratio = 0.95.

An effective data augmentation approach should maintain consistency with the main features, semantic information, and data distribution of the original data after augmentation. In the proposed DAM, the spectrum erasure ratio and the original spectrum retention ratio use the same set of random numbers. Specifically, when the spectrum erasure ratio is large, the erasure intensity used should be smaller; that is, the location to be erased can retain more original information.

Fig. 3 shows the visualization of the augmented sample compared to the original sample at different ratio values. In Fig. 3, when the ratio is very small, such as 0.1, the number of bands processed is small, but the degree of erasure is large. The spectra obtained at this point can be considered as noisy interference, which can be used for network training to increase the noise immunity of the model. As the value of the ratio increases, the number of processed bands increases, but the degree of erasure becomes smaller. For example, with a ratio of 0.95, the generated spectra retain an overall similarity in spectral shapes compared to the original spectra, while exhibiting slight differences that enhance spectral diversity.

D. Convolutional Attention Hybrid Network

The transformer excels in modeling remote global contexts, but it struggles with extracting fine-grained local feature patterns [74]. This is complemented by the ability of CNNs to extract localized information [75]. The hybrid network composed of a transformer and CNN can extract both global and local information [76], [77], [78], [79]. Thus, combining the two can extract subtle differences between spectra and capture relationships between remote spectral bands.

As shown in Fig. 2, the proposed CAN consists of four components: the transformer branch, the CNN branch, the feature fusion, and the detection head. The innovations of the CAN are shown in color in Fig. 2. In the Transformer branch, 1×1 convolutions are used to replace all traditional fully connected layers, and the multilayer perceptron (MLP) consists of two 1×1 convolutions and global pooling to achieve the detection of HSIs of any size; In the CNN branch, an encoder–decoder architecture is used to train the feature extractor, rather than the traditional encoder structure. The encoder–decoder architecture can constrain the network to extract effective features. During training, a modified Davis-Bouldin loss (DBLoss) is used to guide the model in effectively separating anomalies from backgrounds and reducing the distance between backgrounds of different categories to suppress background.

The transformer branch consists of two layers of the encoder. The full connection in the original Transformer structure is replaced by 1×1 convolution, which can enhance the flexibility of the network structure and reduce the complexity of the model. Additionally, two 1×1 convolutions and a global average pooling layer are used to replace the forward propagation layer in the Transformer encoder. This can further integrate global features and reduce the computational complexity of the model. The modified MLP can be expressed as

$$\mathrm{MLP}(\mathbf{x}) = \mathrm{Conv}_{1\times1}\left(\mathrm{GlobalPool}\left(\mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}(\mathbf{x})\right)\right)\right)\right) \tag{5}$$

where ReLU is the activation function, $\mathrm{ReLU}(x) = \max(0, x)$. It will only be activated (output is nonzero) when the input is a positive number, which reduces the amount of calculation of the model. At the same time, the gradient of ReLU is constant to 1 (when $x > 0$), which effectively alleviates the gradient disappearance problem.

To enhance the ability of the CNN branch to extract more representative features, we design it as a network structure with an encoder–decoder. Among them, the encoder consists of a convolutional residual block. Specifically, the input data is residually connected through two 1×3 convolutional layers with the input data through one 1×1 convolutional layer. The decoder includes a transposed convolutional layer that performs spectral dimensional up-sampling to ensure that the number of output bands aligns with the input bands. The CNN branch can be viewed as a process of spectral information reconstruction. Therefore, we use the mean square error loss to guide the model training. To make the reconstructed spectrum values between 0 and 1, we use a sigmoid activation function in the last layer of the decoder.

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The feature fusion consists of concatenation and convolution operations. First, the global features of the transformer branch and the local features of the CNN encoder are concatenated (Concat operation in PyTorch), and then fed into a 1-D convolution with a kernel of 1×3 to obtain global–local fusion features. Next, the fused features are fed into an MLP detection header, which consists of two 1×1 convolutions and a global average pooling, to output feature values. Finally, these values are passed through a sigmoid activation layer to output the probability values of belonging to the anomaly.

Let $\mathbf{F}_g$ represents the global features and $\mathbf{F}_l$ represents the local features. The process of outputting detection results from the extracted features can be expressed in pseudo-code as

$$\mathbf{F}_c = \left[\mathbf{F}_g, \mathbf{F}_l\right] \tag{6}$$

$$\mathbf{F}_f = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 3}\left(\mathbf{F}_c\right)\right)\right) \tag{7}$$

$$D_r = \sigma\left(\mathrm{MLP}\left(\mathbf{F}_f\right)\right) \tag{8}$$

where σ is the Sigmoid function $\sigma(x) = 1/(1 + e^{-x})$. It maps the detection result value to a range of (0, 1), indicating the probability that the pixel is an anomaly. The closer the output result value is to 1, the more likely the pixel is to be detected as an anomaly. Conversely, the closer the output result value is to 0, the more likely the pixel is to be detected as the background.

E. Loss Functions

For the proposed CAN, we utilize a total of two loss functions, each serving a distinct purpose. Specifically, the Davies-Bouldin (DB) loss is used to guide the model in identifying anomalies from the background, while also bringing the different categories of backgrounds closer together. To achieve robust feature extraction in the CNN branch, the mean squared error (MSE) loss is utilized to guide spectral reconstruction. The final loss of the CAN is expressed as follows:

$$\mathcal{L}_{\mathrm{Total}} = \mathcal{L}_{\mathrm{DB}} + \mathcal{L}_{\mathrm{MSE}}. \tag{9}$$

1) Davies-Bouldin Loss: Davies-Bouldin index (DBI) [73] is a clustering evaluation metric used to measure the performance of clustering algorithms. Specifically, it assesses the quality of clustering results by utilizing intracluster cohesion and intercluster separation.

Inspired by this, we utilize the average spectra of the anomaly and background samples as the clustering centers, respectively. The distances between the anomaly and background centers, as well as the distances of the samples to each clustering center, are calculated to guide the network in separating the anomalies from the background. At the same time, a complex background is suppressed by minimizing the distance between different categories of background.

Let $\mathbf{A} \in \mathbb{R}^{N_A \times D}$ denote anomaly samples, and $\mathbf{B} \in \mathbb{R}^{N_B \times D}$ denote background samples. The DB loss is defined as follows:

$$\mathbf{a} = \frac{1}{N_A}\sum \mathbf{A}, \quad \bar{\mathbf{A}} = \mathbf{1}_{N_A} \times \mathbf{a} \tag{10}$$

$$\mathbf{b} = \frac{1}{N_B}\sum \mathbf{B}, \quad \bar{\mathbf{B}} = \mathbf{1}_{N_B} \times \mathbf{b} \tag{11}$$

$$\mathcal{L}_{\mathrm{DB}} = \frac{1}{N_P}\sum \frac{\left\|\bar{\mathbf{B}} - \mathbf{B}\right\|_2 + \left\|\bar{\mathbf{A}} - \mathbf{A}\right\|_2}{\left\|\bar{\mathbf{A}} - \bar{\mathbf{B}}\right\|_2}. \tag{12}$$

TABLE I
DETAILED PARAMETERS OF THE FIVE EXPERIMENTAL DATASETS

Dataset	Size	Anomalies	Number
HYDICE Urban	80×100 × 174	cars and roofs	21
WHU-Hi River	105×168 × 135	plates and panels	36
SIM-GA Scenario	375×450 × 211	tarps,vehicles,panels	135
Hyperion Loukia	249×945 × 175	mining site	67
Zhuhai-1 Ocean	1000×1000 × 32	ships	603

Among them, $\mathbf{a} \in \mathbb{R}^{1\times D}$ denotes the average spectra of the anomaly, $\mathbf{b} \in \mathbb{R}^{1\times D}$ denotes the average spectra of the background. N_A is the number of anomaly pixels, N_B is the number of background pixels, and N_P is the number of anomaly and background sample pairs. To calculate the loss value of the entire batch size sample, we expand the average spectral vector of dimension $\mathbb{R}^{1\times D}$ to a matrix of dimension $\mathbb{R}^{N\times D}$. Based on this, the anomaly average spectrum matrix $\bar{\mathbf{A}}$ can be expressed as $\mathbf{1}_{N_A} \times \mathbf{a}$, and the background average spectrum matrix $\bar{\mathbf{B}}$ can be expressed as $\mathbf{1}_{N_B} \times \mathbf{b}$. Here, $\mathbf{1}_{N_A}$ and $\mathbf{1}_{N_B}$ represent all-one column vectors of dimension N_A and dimension N_B, respectively.

2) Mean Squared Error Loss: MSE loss is often used as a measure of the discrepancy between the predicted values of the model and the true labeled values. It measures the magnitude of the difference by calculating the mean square error and helps optimize the model's parameters. In this work, we utilize this loss function to guide the CNN branch to extract local features. The formula for this loss is as follows:

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N}\sum \left\|\mathbf{X} - \hat{\mathbf{X}}\right\|_2 \tag{13}$$

where $\mathbf{X} \in \mathbb{R}^{N\times D}$ is the input spectrum value, and $\hat{\mathbf{X}} \in \mathbb{R}^{N\times D}$ is the reconstructed spectrum value.

IV. EXPERIMENTAL EVALUATIONS

This section presents an evaluation of the proposed method on five real-world HSIs, with comparisons against classical and state-of-the-art detection approaches.

A. Experimental Setup

1) Datasets: To demonstrate the effectiveness of the proposed method, five real-world HSIs from various sensors and scenes were employed for anomaly detection experiments. Fig. 4 displays the pseudo-color images and anomaly reference images from these datasets. Table I summarizes the anomaly categories, image dimensions (height × width × bands), and the number of anomalies (in pixels) for each dataset.

The HYDICE urban dataset [80] uses a 80 × 100 pixel region, retaining 174 bands after removing those affected by noise and water absorption. Within this area, 21 anomalous pixels, mainly cars and rooftops, are labeled due to their distinct spectral responses from the background.

The WHU-Hi River dataset [81] was collected and shared by the intelligent data extraction, remote sensing analysis, and application (RSIDEA) research group at Wuhan University, China. This data was gathered using a Headwall Nano-Hyperspec sensor mounted on a drone. This dataset has

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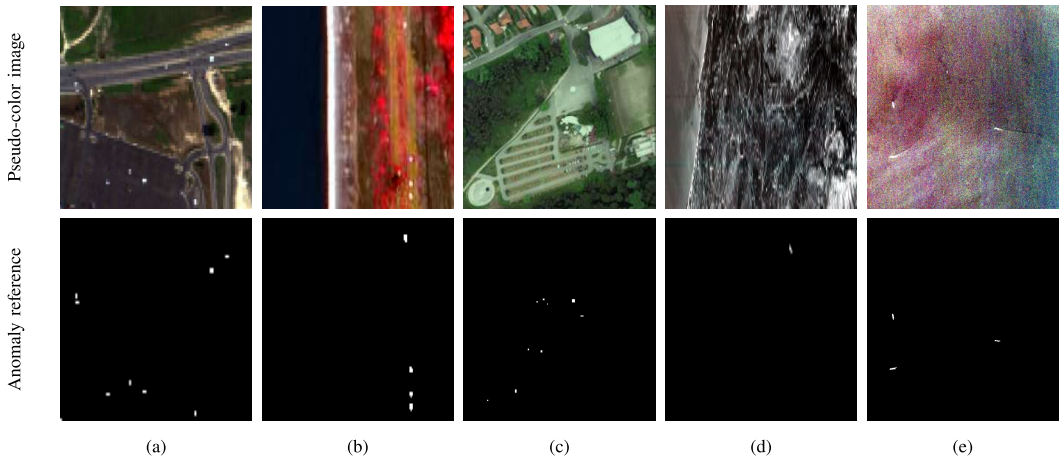


Fig. 4. Pseudo-color images and anomaly reference maps of the experimental datasets. (a) HYDICE. (b) WHU-Hi. (c) SIM-GA. (d) Hyperion. (e) Zhuhai-1.

TABLE II
AUC EVALUATION METRICS INVOLVE DESCRIPTIONS OF ACRONYMS

$AUC_{(D,F)}$	$0 \le AUC_{(D,F)} \le 1$
$AUC_{(D,\tau)}$	$0 \le AUC_{(D,\tau)} \le 1$
$AUC_{(F,\tau)}$	$0 \le AUC_{(F,\tau)} \le 1$
AUC_{ADP}	$0 \le AUC_{ADP} \equiv AUC_{(D,\tau)} \le 1$
AUC_{JAD}	$0 \le AUC_{JAD} = AUC_{(D,F)} + AUC_{ADP} \le 2$
AUC_{BDP}	$0 \le AUC_{BDP} = 1 - AUC_{(F,\tau)} \le 1$
AUC_{JBS}	$0 \le AUC_{JBS} = AUC_{(D,F)} + AUC_{BDP} \le 2$
AUC_{ADBS}	$-1 \le AUC_{ADBS} = AUC_{(D,\tau)} - AUC_{(F,\tau)} \le 1$
AUC_{OAD}	$0 \le AUC_{OAD} = AUC_{ADP} + AUC_{BDP} \le 2$
AUC_{SBPR}	$0 \le AUC_{SBPR} = \frac{AUC_{ADP}}{AUC_{BDP}}$

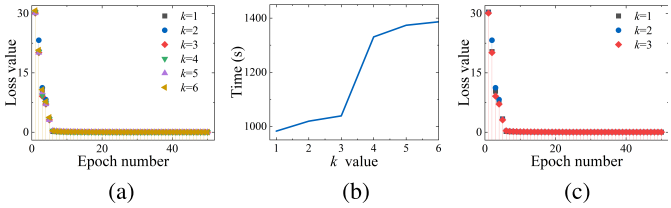


Fig. 5. Relationship between k and loss value. (a) Loss variation with k. (b) Training time (in seconds) for different k values, where higher k increases cost but has little effect on convergence. $k = 4$ sharply increases training time, while lower k values reduce it. (c) Loss convergence for k between 1 and 3, with $k = 3$ (red symbol) converging fastest.

dimensions of 105×168 pixels and includes 135 bands. Plastic sheets and gray panels are classified as anomalies and occupy 36 pixels.

The SIM-GA scene dataset [82] originates from the shared α scene of the "Viareggio 2013 experiment collected by a push-broom hyperspectral SIM-GA sensor. This dataset has dimensions of 375×450 pixels and includes 211 bands. The background consists of a natural environment that features a river, where artificially placed tarpaulins, vehicles, and panels are considered anomalies, which occupy 135 pixels.

The Hyperion Loukia dataset [83] was collected by the Hyperion sensor, with spatial dimensions of 249×945 pixels and 175 bands. Mineral mining sites, which are not visually prominent, are classified as anomalies, occupying 67 pixels.

The Zhuhai-1 Ocean dataset, based on an actual HSI from the Zhuhai-1 satellite, contains 1000×1000 pixels across 32 bands. Ships—designated as anomalies— occupy 603 pixels. The presence of several anomalous-like targets and background similarities significantly increases the difficulty of accurate ship detection.

2) Evaluation Metrics: The effectiveness of a hyperspectral anomaly detector depends on two key issues: anomaly detectability and background suppression. We evaluated the experimental results using a set of metrics specifically designed for assessing the performance of HAD methods [84], [85]. These metrics include the effectiveness of the detector (P_D, P_F), anomaly detection probability (ADP), joint target detectability (JAD), background detection probability (BDP), joint background suppressibility (JBS), joint anomaly detectability and background suppressibility (ADBS), concerning overall anomaly detection (OAD), and signal-to-background probability ratio (SBPR). Table II summarizes the calculation instructions of these evaluation indicators. Among them, P_D represents the detection probability, P_F represents the false alarm probability, and τ represents the threshold.

3) Implementation Details: The proposed model is trained using the HAD100 dataset. For more details about this dataset, refer to the URL: https://zhaoxuli123.github.io/HAD100/. The proposed method initially classifies the input dataset into potential anomalies and backgrounds through an adaptive sparsity module in the training phase. The network then learns the differences between them to identify the anomalies. Therefore, we used HSIs containing anomalies in the AVIRIS-NG dataset for training.

Specifically, the model training epoch size is 50, the initial learning rate is 0.01, and the Adam optimizer is used. To alleviate overfitting, the weight decay is 0.001. In the experiment, the hidden dimension of the network is set to 16, which increases the possibility of successful training and reduces the computational resource requirements. The detailed structure of the proposed network is given in Section II.

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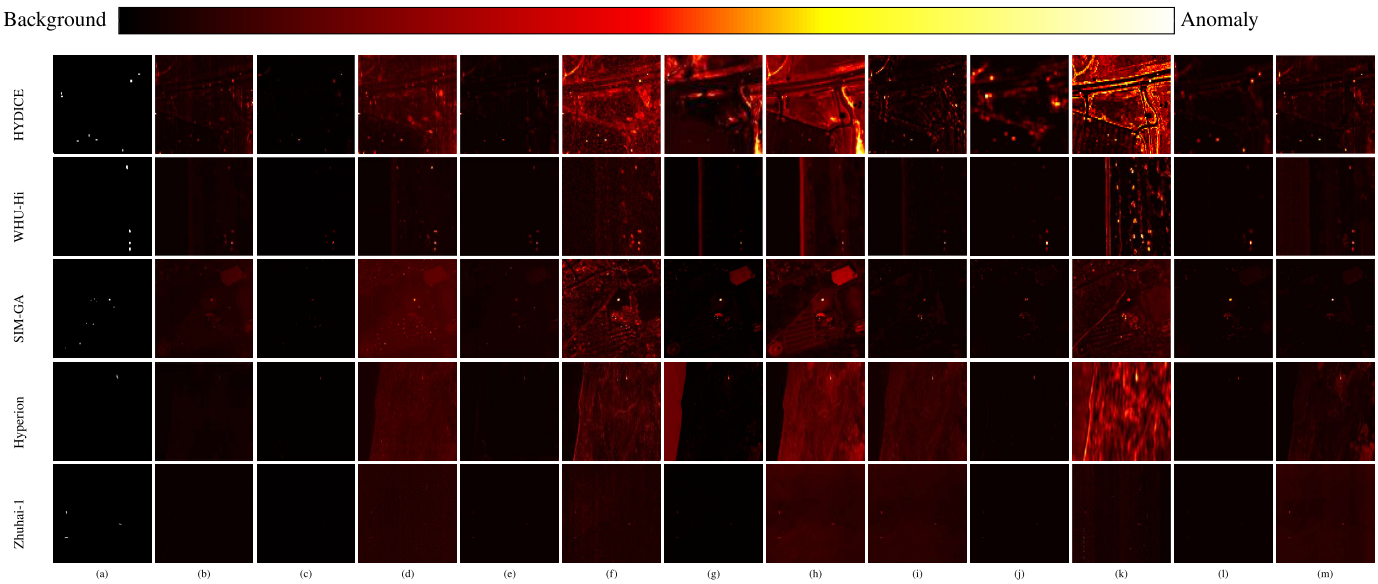


Fig. 6. Anomaly reference maps and detection maps of the compared algorithms on five datasets. (a) Anomaly reference. (b) RX. (c) SSRX. (d) CRD. (e) LSMAD. (f) PCA-TLRSR. (g) RGAE. (h) Auto-AD. (i) PDBSNet. (j) GT-HAD. (k) AUD-Net. (l) FEN. (m) AdaptHAD.

In the ASM, k represents the order of the norm. It can be seen from (1) that the larger the value of k, the greater the consumption of computer resources. Considering the computational cost, we only show the impact on model training when k values are from 1 to 6. Fig. 5 shows the relationship between parameter k and the convergence of the model loss function. It can be seen that when the value of parameter k is 3, the model is easier to train successfully.

To demonstrate the effectiveness of the proposed method, we select 11 methods with excellent detection performance: RX [13], SSRX [17], CRD [25], LSMAD [28], PCA-TLRSR [30], RGAE [46], Auto-AD [38], PDBSNet [39], GT-HAD [54], AUD-Net [62], and FEN [63]. These methods cover both classic and state-of-the-art approaches, including those statistics-based, representation learning-based, and deep learning-based methods.

To ensure the fairness of the experimental comparison, the parameter settings of each method are configured according to its original literature. The relevant details are shown below. In the experiments, the traditional RX method has no parameters to set. The parameters in the SSRX, LSMAD, PCA-TLRSR, GT-HAD, and Auto-AD methods were set according to the original papers.

For the CRD method, the window size $(w_{\text{out}}, w_{\text{in}})$ was set to $(19, 15)$ on the HYDICE, SIM-GA, and Hyperion datasets, and $(7, 5)$ on the WHU-Hi and Zhuhai-1 datasets, with the regularization parameter λ fixed at 10^{-6}. In the RGAE method, the trade-off parameter λ, the number of superpixels S, and the hidden-layer dimension n_{hid} were set to $(10^{-2}, 150, 100)$ for all datasets. The AUD-Net method requires the setting of double window sizes. In this work, we selected the optimal combination of inner and outer window sizes based on the recommendations provided in the original article and trained it on the WHU Longkou dataset [86]. Specifically, the Hyperion dataset is $(7, 15)$, and the Zhuhai-1 dataset is $(5, 23)$, the parameters of the remaining datasets are set to $(1, 5)$. For the PDBSNet method, the training and testing factors were set equal, and the factor with the best detection performance was chosen for each dataset: 3 for WHU-Hi and Hyperion, 4 for HYDICE, and 5 for SIM-GA and Zhuhai-1. Finally, the FEN method configures its channel value according to the number of spectral bands, with values of 100 for HYDICE, WHU-Hi, and Hyperion; 200 for SIM-GA; and 30 for Zhuhai-1.

B. Detection Performance

1) Qualitative Analysis: Fig. 6 shows the qualitative detection results of all methods on five experimental datasets. In Fig. 6, we use a color scale bar that has gradients in the order of black, red, yellow, and white to represent the probability that a pixel is an anomaly. The color gradient order corresponds to the probability value from small to large (the value range is 0–1). Specifically, the pixel color in the heat map is closer to white, the greater the possibility that the pixel is an anomaly. On the contrary, the closer to black, the more likely the pixel belongs to the background. In an ideal situation with excellent detection performance, the anomalies are all white and the background is all black. As shown in Fig. 6(a), the distribution of the actual locations of anomalies and background in the experimental dataset.

As shown in Fig. 6, in terms of anomaly detectability, methods such as AdaptHAD, FEN, Auto-AD, LSMAD, and CRD are capable of identifying anomalies to varying extents. Among them, AdaptHAD stands out by consistently detecting all anomaly targets across different datasets. For instance, in the WHU-Hi dataset, which contains four anomaly objects, AdaptHAD successfully identifies all of them, highlighting its superior anomaly detection capability.

2) Quantitative Analysis: Fig. 7 shows the 3-D ROC curves along with their generated three 2-D ROC curves for all compared methods on the experimental dataset. Fig. 7(a) shows the ROC curve generated in (P_D, P_F, τ) 3-D space. With

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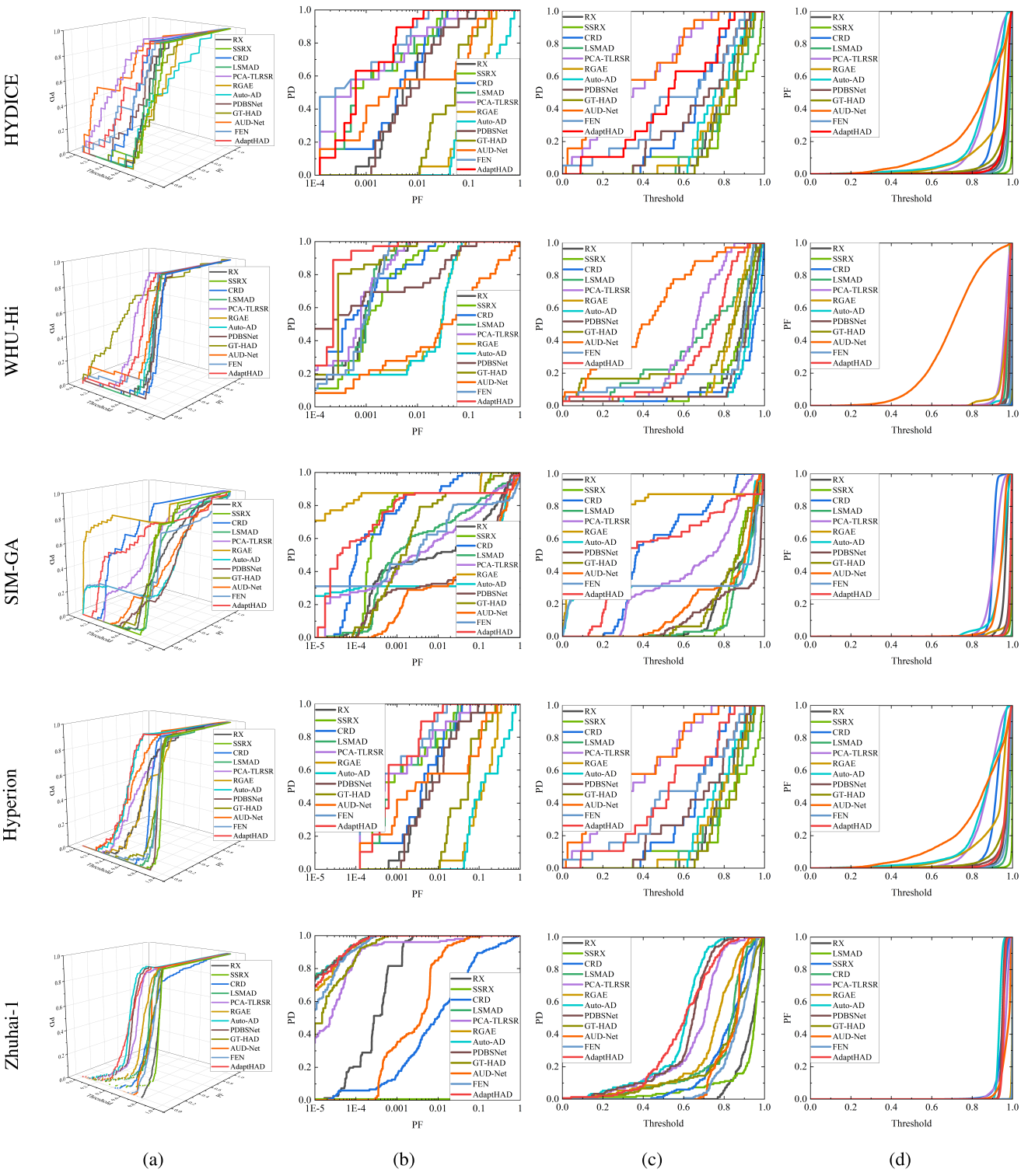


Fig. 7. Three-dimensional ROC maps of the comparison methods on the five datasets. (a) 3D ROC curve. (b) 2D ROC curve of (P_D, P_F). (c) 2D ROC curve of(P_D, τ). (d) 2D ROC curve of(P_F, τ).

such a 3-D ROC curve, three 2-D ROC curves can also be generated, namely the 2-D ROC curve of $\mathrm{AUC}_{(D,F)}$, the 2-D ROC curve of $\mathrm{AUC}_{(D,\tau)}$, and the 2-D ROC curve $\mathrm{AUC}_{(F,\tau)}$, corresponding to Fig. 7(b)–(d), respectively.

The red curve in Fig. 7 represents the proposed method. Fig. 7(b) exhibits that when the PF value is less than 0.001, the red curve is at the top, indicating that our method performs better than other methods. Within a smaller threshold range, the red curve in Fig. 7(c) is closer to the upper left corner on different datasets, indicating that the anomaly highlighting effect of our method is obvious and stable. Fig. 7(d) shows that the red curve is close to the lower right on different

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TABLE III

COMPARISON OF DIFFERENT METHODS ON FIVE HYPERSPECTRAL DATASETS. THE TOP THREE METRICS IN EACH COLUMN ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Dataset	Method	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{JAD}	AUC_{OAD}	AUC_{BDP}	AUC_{JBS}	AUC_{ADBS}	AUC_{SBPR}
HYDICE	RX	0.9824	0.1948	1.1772	1.1699	0.9751	1.9575	0.1699	0.1998
	SSRX	0.9937	0.1342	1.1279	1.1331	0.9988	1.9925	0.1331	0.1344
	CRD	0.9906	0.3304	1.3210	1.2689	0.9384	1.9290	0.2689	0.3521
	LSMAD	0.9940	0.1823	1.1764	1.1733	0.9909	1.9850	0.1733	0.1840
	PCA-TLRSR	0.9928	0.5963	1.5892	1.4891	0.8927	1.8856	0.4891	0.6679
	RGAE	0.8490	0.2069	1.0559	1.1352	0.9283	1.7774	0.1352	0.2228
	Auto-AD	0.7502	0.2254	0.9756	1.1008	0.8754	1.6256	0.1008	0.2574
	PDBSNet	0.9832	0.3049	1.2881	1.2925	0.9876	1.9708	0.2925	0.3087
	GT-HAD	0.9336	0.1683	1.1020	1.1433	0.9749	1.9086	0.1433	0.1726
	AUD-Net	0.9481	0.6604	1.6086	1.5624	0.9020	1.8501	0.5624	0.7322
	FEN	0.9973	0.3546	1.3520	1.3395	0.9848	1.9822	0.3395	0.3601
	AdaptHAD	0.9977	0.4183	1.4161	1.3938	0.9755	1.9733	0.3938	0.4288
WHU-Hi	RX	0.9987	0.1278	1.1265	1.1150	0.9872	1.9860	0.1150	0.1294
	SSRX	0.9965	0.0738	1.0703	1.0735	0.9996	1.9962	0.0735	0.0738
	CRD	0.9970	0.2808	1.2779	1.2616	0.9807	1.9778	0.2616	0.2863
	LSMAD	0.9987	0.1217	1.1205	1.1199	0.9981	1.9969	0.1199	0.1219
	PCA-TLRSR	0.9833	0.3092	1.2926	1.2140	0.9048	1.8882	0.2140	0.3417
	RGAE	0.9721	0.0827	1.0549	1.0763	0.9935	1.9657	0.0763	0.0833
	Auto-AD	0.9724	0.1683	1.1408	1.1364	0.9680	1.9404	0.1364	0.1739
	PDBSNet	0.9884	0.0870	1.0755	1.0768	0.9897	1.9781	0.0768	0.0879
	GT-HAD	0.9992	0.1612	1.1604	1.1598	0.9986	1.9978	0.1598	0.1614
	AUD-Net	0.9311	0.6818	1.6129	1.2321	0.5502	1.4814	0.2321	1.2390
	FEN	0.9989	0.1799	1.1788	1.1761	0.9962	1.9951	0.1761	0.1806
	AdaptHAD	0.9996	0.2353	1.2350	1.2097	0.9743	1.9740	0.2097	0.2415
SIM-GA	RX	0.8767	0.2116	1.0883	1.1746	0.9629	1.8397	0.1746	0.2197
	SSRX	0.9791	0.1231	1.1023	1.1227	0.9996	1.9787	0.1227	0.1232
	CRD	0.9969	0.5132	1.5101	1.4156	0.9024	1.8993	0.4156	0.5686
	LSMAD	0.8786	0.1329	1.0116	1.1095	0.9765	1.8552	0.1095	0.1361
	PCA-TLRSR	0.7420	0.3517	1.0938	1.2150	0.8632	1.6053	0.2150	0.4075
	RGAE	0.9863	0.7473	1.7336	1.7351	0.9878	1.9741	0.7351	0.7565
	Auto-AD	0.7529	0.3207	1.0737	1.2597	0.9389	1.6919	0.2597	0.3416
	PDBSNet	0.8130	0.1056	0.9187	1.0928	0.9872	1.8002	0.0928	0.1070
	GT-HAD	0.9732	0.1559	1.1292	1.1498	0.9939	1.9672	0.1498	0.1569
	AUD-Net	0.6284	0.6679	1.2963	1.0403	0.3724	1.0008	0.0403	1.7934
	FEN	0.8503	0.1767	1.0270	1.1657	0.9890	1.8393	0.1657	0.1786
	AdaptHAD	0.9141	0.6090	1.5232	1.5973	0.9882	1.9024	0.5973	0.6163
Hyperion	RX	0.9902	0.0606	1.0508	1.0443	0.9837	1.9740	0.0443	0.0616
	SSRX	0.9946	0.0271	1.0218	1.0266	0.9995	1.9941	0.0266	0.0271
	CRD	0.9784	0.1567	1.1352	1.0771	0.9203	1.8987	0.0771	0.1703
	LSMAD	0.9856	0.0945	1.0802	1.0814	0.9868	1.9725	0.0814	0.0958
	PCA-TLRSR	0.9565	0.3363	1.2929	1.2677	0.9313	1.8878	0.2677	0.3611
	RGAE	0.9474	0.2474	1.1948	1.2130	0.9656	1.9130	0.2130	0.2562
	Auto-AD	0.9988	0.4260	1.4249	1.3202	0.8942	1.8930	0.3202	0.4764
	PDBSNet	0.9690	0.2604	1.2294	1.1965	0.9361	1.9052	0.1965	0.2781
	GT-HAD	0.9958	0.1042	1.1000	1.1013	0.9971	1.9929	0.1013	0.1045
	AUD-Net	0.8492	0.5673	1.4166	1.0517	0.4844	1.3337	0.0517	1.1711
	FEN	0.9993	0.0829	1.0822	1.0799	0.9970	1.9964	0.07999	0.0831
	AdaptHAD	0.9997	0.4556	1.4553	1.3869	0.9313	1.9310	0.3869	0.4892
Zhuhai-1	RX	0.9994	0.0754	1.0749	1.0747	0.9992	1.9987	0.0747	0.0755
	SSRX	0.9999	0.0791	1.0790	1.0790	0.9998	1.9998	0.0790	0.0791
	CRD	0.9415	0.1866	1.1282	1.1330	0.9463	1.8879	0.1330	0.1972
	LSMAD	0.9999	0.1785	1.1785	1.1773	0.9988	1.9988	0.1773	0.1787
	PCA-TLRSR	0.9987	0.3429	1.3416	1.3111	0.9683	1.9670	0.3111	0.3541
	RGAE	0.9999	0.2309	1.2309	1.2275	0.9966	1.9965	0.2275	0.2317
	Auto-AD	0.9999	0.3917	1.3916	1.3262	0.9345	1.9345	0.3262	0.4191
	PDBSNet	0.9998	0.3586	1.3584	1.3069	0.9483	1.9482	0.3069	0.3781
	GT-HAD	0.9999	0.1722	1.1721	1.1717	0.9994	1.9994	0.1717	0.1723
	AUD-Net	0.9024	0.6761	1.5786	1.2437	0.5675	1.4700	0.2437	1.1913
	FEN	0.9999	0.1223	1.1223	1.1207	0.9984	1.9983	0.1207	0.1225
	AdaptHAD	0.9999	0.3617	1.3617	1.3166	0.9548	1.9548	0.3166	0.3788

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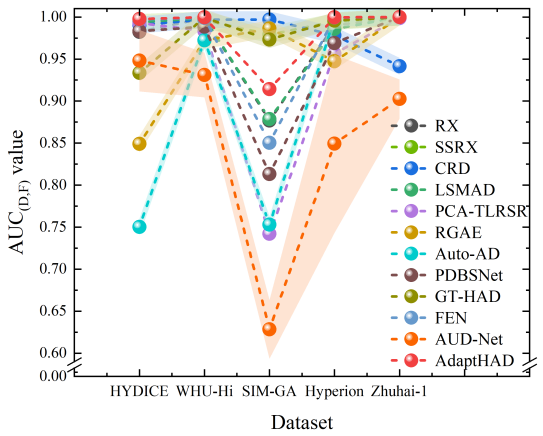


Fig. 8. Fluctuation range of $AUC_{(D,F)}$ values of each method on different datasets.

datasets, indicating that the background suppression of this method is effective and stable. This may be attributed to the operation of the proposed method to minimize the differences between different classes of backgrounds in HSIs during model training.

To further quantify the detection performance and stability between the comparison methods, we conducted ten experiments for each method and calculated their mean and standard deviation on multiple AUC metrics. According to the calculation results, we found that the standard deviations of most comparison methods are very small, and only a few methods have relatively large standard deviations. Therefore, we recorded the average value of ten runs in Table III to compare and analyze the experimental results more clearly. In addition, to illustrate the stability of the model's performance and considering that $AUC_{(D,F)}$ is a widely recognized evaluation metric for HAD, we plotted the mean and standard deviation of $AUC_{(D,F)}$ in Fig. 8.

Table III summarizes their series of mean AUC estimates derived from 3-D ROC. Specifically, $AUC_{(D,F)}$ measures the overall detection effectiveness; AUC_{ADP}, AUC_{JAD}, and AUC_{OAD} reflect the detector's anomaly detection capabilities; AUC_{BDP} and AUC_{JBS} evaluate background suppression performance; AUC_{ADBS} captures the joint performance in anomaly detection and background suppression; and AUC_{SBPR} indicates the signal-to-background performance ratio.

In terms of anomaly detection performance, AdaptHAD and AUD-Net maintained top three rankings across multiple datasets, suggesting their superior generalization capabilities. The PCA-TLRSR also demonstrated good detection performance on the first two datasets, with Auto-AD performing better on the larger Hyperion and Zhuhai-1 datasets than on the smaller datasets. The RGAE performed better than other methods on the SIM-GA dataset, which has multiple anomalies. Regarding background suppression, performance among different methods was inconsistent. Nevertheless, the AdaptHAD and PCA-TLRSR generally achieved higher and more stable rankings on most datasets, demonstrating their better adaptability to different background types. The SSRX achieved good performance on background detection probability and joint background suppressibility across multiple datasets. AUD-Net performed well on the smaller HYDICE and WHU-Hi datasets. In contrast, Auto-AD performed better on the larger Hyperion and Zhuhai-1 datasets. In terms of signal-to-noise ratio, both AdaptHAD and AUD-Net performed well and were stable. PCA-TLRSR also performs well on the first four datasets, but its performance drops slightly on the larger Hyperion and Zhuhai-1 datasets.

Fig. 8 illustrates the fluctuation range of the $AUC_{(D,F)}$ values for various methods on different datasets. The horizontal axis represents the five experimental datasets, while the vertical axis denotes the average AUC values obtained from ten independent runs. The shaded area around each curve indicates the standard deviation range of the corresponding method. The red markers highlight the proposed method, which demonstrates relatively stable performance across all datasets. In contrast, the AUD-Net method shows a wider shaded area, indicating that the stability of this method is relatively poor.

C. Computational Efficiency

Table IV presents the running time (in seconds), computational complexity (Giga MACs), and the number of model parameters (in millions, M) for the comparison methods on the experimental datasets. Traditional methods such as RX, LSMAD, and PCA-TLRSR do not require training; therefore, only their detection time and computational complexity are reported, while the number of model parameters is marked as "–." For deep learning-based methods, the reported time includes both training and detection. Furthermore, all experiments were run on the same device.

Notably, the proposed AdaptHAD method and AUD-Net require training only once, resulting in a consistent number of model parameters across different datasets. In contrast, other deep models must be retrained for each dataset, leading to variations in their parameter counts. In general, AdaptHAD achieves the lowest number of model parameters, making it more efficient in terms of computational resource usage and less dependent on large-scale training samples.

D. Ablation Study

To demonstrate the effectiveness of the proposed method, we utilize the WHU-Hi dataset as a case study. Since $AUC_{(D,F)}$, $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$ are recognized quantitative metrics in the field of HAD, we use them to measure the ablation experimental performance.

Specifically, Table V shows the ablation experimental results of the three modules of ASM, data augment method (DAM), and CAN in the proposed method, where the bold font indicates the best performance under this metric. From the results, it can be seen that removing the ASM or DAM module will lead to a significant drop in performance, indicating that the two play a key role in the model's effect. In contrast, the best performance can be obtained when the three are introduced at the same time, which verifies the synergistic gain between the modules and the effectiveness of the overall design.

As shown in Fig. 9, the darker the color, the more likely it is the background, and the brighter the color, the more

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TABLE IV
COMPUTATIONAL EFFICIENCY OF COMPARISON METHODS (UNIT: SECOND)

Method	HYDICE	WHU-Hi	SIM-GA	Hyperion	Zhuhai-1	MACs (G)	Params (M)
RX	0.05	0.14	3.26	2.05	1.68	–	–
SSRX	1.76	3.88	768.72	53.24	63.94	–	–
CRD	8.84	2.99	358.69	302.81	1059.06	–	–
LSMAD	5.23	10.63	521.09	225.09	151.00	–	–
PCA-TLRSR	6.30	17.54	185.54	683.92	1544.22	–	–
RGAE	21.96	185.85	3779.32	2005.80	15757.45	–	–
Auto-AD	61.60	67.10	713.19	353.39	4783.32	132.6291±191.3261	3.2236±0.2354
PDBSNet	69.45	363.23	1524.39	1240.84	2203.34	128.9375±203.1893	0.5479±0.0157
GT-HAD	51.10	57.82	829.67	873.68	1299.06	0.0008±0.0001	0.0105±0.0001
AUD-Net	72857.36	72854.31	73032.81	73043.83	73798.64	6.375±0.0	2.37±0.0
FEN	1802.27	1802.99	1806.58	1805.36	1808.13	30.9928±0.4367	83.8932±0.0621
AdaptHAD	1039.41	1039.56	1046.63	1045.28	1058.65	0.0008±0.0	0.0057±0.0

TABLE V
EFFECT OF MODULE COMPONENTS ON THE PROPOSED FRAMEWORK

ASM	DAM	CAN	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$
	✓	✓	0.0261	0.4517	0.8296
✓		✓	0.0263	0.4398	0.8217
✓	✓		0.9541	**0.4580**	0.1586
✓	✓	✓	**0.9996**	0.2353	**0.0256**



(a) (b) (c)

Fig. 9. Detection map on the WHU-Hi dataset. (a) Transformer. (b) CNN. (c) CAN.

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT DATA AUGMENTATION STRATEGIES

	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$
Flipping	0.0321	0.5395	0.8734
Spectral mixing	0.0269	0.4590	0.8257
Random noise addition	0.9751	**0.5813**	0.2408
DAM (ours)	**0.9996**	0.2353	**0.0256**

likely it is an anomaly. Fig. 9(a) The Transformer performs well in highlighting anomalies in the detection image, but the background suppression effect is poor. Fig. 9(b) CNN shows excellent background suppression ability, but unfortunately, the anomaly is also suppressed at the same time. In contrast, (c) CAN can effectively suppress the background and highlight the anomaly.

Flipping, spectral mixing, and random noise addition are commonly used HSI DAMs [87]. To demonstrate the effectiveness of the proposed method, we compare and analyze it with the above typical augmentation methods. All experiments are uniformly conducted on the WHU-Hi dataset. The results are shown in Table VI. As shown in the table, the DAM strategy achieves the highest $AUC_{(D,F)}$ value, demonstrating superior performance and stronger discriminative capability in HAD. In contrast, the flip and spectral mixing augmentation strategies achieve lower scores, suggesting that these methods are less effective in enhancing model performance and may be suboptimal for data augmentation in this scenario.

V. CONCLUSION

In this work, an adaptive one-step hybrid network is proposed for HAD, which solves the problem of limited input image dimensionality. In addition, this method does not require retraining or parameter fine-tuning when detecting different HSIs. This reduces the waste of resources caused by repeated training. Experimental analysis shows that the proposed method is effective and stable in different datasets. Our future work will focus on modeling the correlations between samples across batches for effective multiclass anomaly detection.

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